

GRAND MONOGRAPH: TECHNICAL DRAFT STATUS & EPISTEMIC SCOPE

Document Classification: Internal Technical Draft / Working Compendium

Audit Reference: THEORY_CONSISTENCY_AUDIT.md & VERIFICATION_RUN_002

Status Statement: Ready for internal compilation; canonical MAP benchmark reported with full objective-function disclosure; independent rerun pending artifact inspection.

Date: October 9, 2026

Epistemic Boundary Ledger & Multi-Tier Certification

This compendium presents the unified compilation of the Conformal Sovereign Action, galactic rotation curve dynamics, high-redshift observations, and cosmological horizon physics. In accordance with the Sovereign Epistemic Covenant, all mathematical and empirical results are strictly stratified:

- **[P] Proved / Kernel Verified:** Mathematical propositions and algebraic identities mechanically checked by Lean 4 / Mathlib (all 6 formal modules build with exit code 0, depending solely on standard foundationals `[propext, Classical.choice, Quot.sound]`).
- **[D] Direct Empirical / Computed:** Quantities directly computed from raw, hashed astronomical databases (e.g., SPARC 175-galaxy rotation curves, SHA-256: `e76e6752164b...`).
- **[C] Cited External Literature:** Standard literature values and empirical baselines (e.g., Planck 2018 parameters, Lelli et al. 2016).
- **[O] Open Problem / Conjectured Boundary:** Theoretical bridge hypotheses, cosmological normalizations without closed variational derivations, and effective optical couplings.

1. Verified Mechanical Core vs. Physical Open Problems

Sector / Claim	Status	Scope & Boundary Definition
Lean 4 Module Suite	[P] Verified	Formally verifies mathematical theorems from encoded definitions.
SPARC Tier 0 (prior-mean M/L , $a_0 = \frac{cH_0}{2\pi}$ declared)	[D] Computed	Median $\chi^2_{\text{data}}/N_g = 11.08$ vs MOND 9.93 (171 galaxies, <code>PARAMETER_LEDGER.json</code>). Two irreducible inputs: the a_0 scale and the μ choice.
SPARC strict unit- M/L check	[D] Computed	Median $\chi^2_{\text{data}}/N_g = 15.01$, aggregate $\sum \chi^2_{\text{data}}/\text{dof} = 69.07$ (3,391/3,391, <code>SPARC.175.summary.json</code>).
SPARC grid nuisance fit	[D] Computed	Median $\chi^2_{\text{data}}/N_g = 3.06$, aggregate $\sum \chi^2_{\text{data}}/\text{dof} = 7.78$ (3,391/3,009, $N_{\text{par}} = 382$, Gaussian priors).
Baryons-Only Baseline (Unit M/L)	[D] Computed	Median $\chi^2_{\text{data}}/N_g = 51.58$, aggregate $\sum \chi^2_{\text{data}}/\text{dof} = 204.66$.
Baryons-Only Baseline (prior-mean M/L)	[D] Computed	Median $\chi^2_{\text{data}}/N_g = 85.23$, aggregate $\sum \chi^2_{\text{data}}/\text{dof} = 406.49$.
Action $\rightarrow \mu_{\text{std}}(x)$ Bridge	[O] Open	Direct variation yields $\text{arcsinh}(x)$; the closure to $\mu_{\text{std}} = x/\sqrt{1+x^2}$ is open.
$a_0 = \frac{cH_0}{2\pi}$ Normalization	[O] Open	$T_U = T_{\text{GH}}$ yields $a = cH_0$; the $1/(2\pi)$ divisor is an open residue.
Optical Metric & Redshift Jacobian	[O] Open	Effective disformal couplings; Maxwell/Poynting derivations pending.

Statistical Note: The displayed aggregate 7.93 is $\sum \chi^2_{\text{data}}/3009$, using nominal data-residual dof ($N_{\text{data}} - N_{\text{fitted}} = 3391 - 382 = 3009$). Because parameters are MAP-regularized by informative priors, this denominator is a reporting convention rather than a calibrated posterior goodness-of-fit distribution.

2. Official Gate Verdicts

Certification Gate	Verdict	Rationale
Mechanical Build / Compilation	READY	All 6 Lean modules compile cleanly (<code>exit 0</code>), TeX sources sound.
Internal Technical Draft	READY	Complete transparency, canonical MAP single-run parameterization recorded.
Independent Reproducibility	PENDING	Executable Run 002 logs, data manifest, and residual CSVs archived for inspection.
External Referee Submission	HOLD	Action $\rightarrow \mu(x)$ bridge requires explicit variational closure; a_0 is an open gap.
Unified Theory Validation Claim	HOLD	Active theoretical program with well-defined open problems.